Inequality in a Potential Equalizer of Opportunity*

Variation in the Accessibility and Net Benefits of English-Medium Instruction in India

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This document contains the Introduction and Sections 4.1, 4.2, and 4.3 of the titular paper. The project may have been updated since this file was generated; the most up-to-date versions of this document, the full paper, the replication package, and other application samples are available online for those interested. This PDF can be generated by running `make samples` or `scripts/run_full_build.sh` from the repo root.

Abstract

English-medium instruction (EMI), or the teaching of school subjects in English, is often viewed as a potential palliative for the intergenerational rigidity of socioeconomic status in India, where English skills command substantial wage premia. I study whether the conditions governing households' access to and potential net gains from EMI are themselves unequally distributed. I organize my analysis around three such conditions: inherited linguistic costs to acquiring English, access to English-medium schooling, and the local economic setting in which English skills may be rewarded. By harmonizing district data on languages, schooling, consumption, migration, human capital, and sectoral employment, I find inequality across all three. Inherited linguistic conditions are highly persistent and predict future human capital gains, financial asset access, and structural change, but their relationship with language acquisition behavior varies across institutional settings. EMI exposure and provision are strongly organized by schooling system and sharply stratified along socioeconomic lines, even within the same local education market. Consumption growth is more associated with EMI enrollment than enrollment alone, and the strength of their association increases in pre-existing human capital and urbanization. In sum, a high-return educational opportunity need not equalize opportunity when access to it and the resources which help benefit from it are unequally distributed.

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1 Introduction

In an Indian economy characterized by rapid GDP growth alongside wage stagnation, profuse informality, and high rates of both youth unemployment and underemployment (International Labour Organization 2024), what can a household do to improve the future prospects of their children? The history of economics highlights *education* as one answer (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India highlights another: *English*. In a context of extraordinary linguistic diversity (Gurevich et al. 2025) and thus extraordinary communication and coordination costs (Nakagawa and Sugasawa 2021), English has come to serve as a primary domestic lingua franca (Clingingsmith 2014). In its capacity as a global lingua franca (Melitz 2016), meanwhile, English opens pathways both internationally—e.g., to education at prestigious universities or employment at multinational firms (Ramanathan 2014; Bedi 2019, 32–33)—and at home. The domestic expansion of internationally traded services and information technology, in particular, has led to a flurry of jobs whose higher socioeconomic returns (Chamarbagwala and Sharma 2011; Smokotin et al. 2014) are largely accessible only to those who can use English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study corroborates the notion that English expands economic mobility (Itani et al. 2015). Using nationally representative household data, Azam et al. (2013) find that English fluency was, at least up until 2005, associated with a 34% hourly wage premium for men and a 22% premium for women; even limited English proficiency was associated with premia of 13% and 10%, respectively. Refeque and Azad (2022) find similar differentials extending through 2011–12. Evidence for causality in this association comes from Chakraborty and Bakshi (2016), who interpret West Bengal's 1983–2004 ban on English teaching in public primary schools as a natural experiment where cohorts more *exposed* to the policy earned lower subsequent wages. "Language skills are human capital," after all (Chiswick and Miller 1995, 4), and in a context like India especially, "English-language skills [specifically] are a form of human capital" (Azam et al. 2013, 344).

It is thus by combining our two answers—education and English—that we can begin to understand the demand for *English-medium instruction (EMI)*, or for the teaching of all school subjects in English, currently seen in India (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhay and Roy 2017). For its potential to expand equality of opportunity, EMI has been touted as a tool of social development (National Knowledge Commission 2009) and described as a route to escape caste (Mathur 2013; Mathew and Srivastava 2011; Vij 2020) alongside the rigid wage, employment, and occupation distributions it enforces across generations (Madheswaran and Singhari 2016; Munshi and Rosenzweig 2006).

But it is ultimately households who decide whether or not a child attends EMI, and the determinants of their choice—the costs they face, the benefits they expect, and the forces constraining the choice set in front of them—are affected by factors which are not just national and historical, but also specific and local. My focus is on whether

such factors are associated with the same inequalities that English skills are often expected to help households escape. I study inequalities which exist along dimensions such as social group, religion, parental education, household resources, and urban or rural residence. If the ability to access or benefit from EMI is systematically related to these characteristics, even within the same local education market, then we have evidence that a high economic payoff to English skills alone is not enough for EMI to become an equalizer of opportunity.

Though I never estimate a structural model of household choice, their investment problem still provides a helpful organizing framework. I will first study a type of cost whose variation is predetermined relative to the institutional and economic factors that otherwise affect the household's problem: **inherited linguistic costs to learning English**. These will reflect historically persistent features of the local linguistic environment (see Rowe and Weisleder 2020) which affect the opportunity cost of learning English over other domestic bridge languages, namely Hindi. This work, detailed in Section 3, owes much credit to Shastry (2012); in line with her methodology, I create several different measures of local linguistic distance from *Hindi* to capture this opportunity cost. In Section 4 I analyze the **accessibility of English-medium instruction** through enrollment, various measures of EMI exposure, and private school management. Finally, in Section 5 I study potential determinants of **expected returns to English skills**. My goal here is to find evidence on the value of English skills in the local labor market, with the assumption that households incorporate the local history of such returns into their beliefs regarding English, akin to households having adaptive expectations. I ask whether EMI exposure predicts later district consumption and whether that association is stronger where pre-existing human capital and urbanization are greater. I then examine whether inherited linguistic conditions accompany human-capital growth and skilled in-migration, as well as changes in financial access and economic structure.

The local linguistic environment, the local education market, and the local labor market—all three of the factors I focus on are studied locally. These choices are justifiable: the linguistic conditions which households pass down to their children are highly persistent at the local level, local schooling conditions constrain the menu of educational opportunities households choose from, and local economic conditions plausibly cause households to update their beliefs on the economic returns to learning English. I will use a household's district to proxy for their 'local' environment.[1] Though the district is the most granular unit at which I can harmonize the longitudinal data sources I rely on, it must be noted that interdistrict migration would confound and attenuate any association between the characteristics of one district's education market and its future economic outcomes. Older studies found interdistrict migration rates (and particularly rates of economically-motivated interdistrict migration) to be so low even after the expansion of IT and services in the 1990s (Topalova 2005; National Statistical Office 2021) that they "assume[d] that district of current residence (or of employment) of an individual is approximately

[1]A district in India is a political unit akin to a county in the United States, albeit with an average population of around 2 million. Table 1 contains summary statistics derived from the distribution of district populations.

the same as the schooling district" (Chakraborty and Bakshi 2016, 6). More recent work, however, complicates (Mahapatro 2020; Rajan et al. 2020) and even outright repudiates (Lucas 2021) this notion. These studies emphasize the role of the IT expansion in driving interdistrict migration (Economic Divison 2015; Ghose 2026), and my own results associate potential determinants of household investment into EMI with the skill composition of recent work migrants. I return to those results in Section B. Regardless, it is evident that district outcomes are not an aggregation of individual returns to EMI, but a function of places and their changing populations.

The results show that the conditions surrounding English-skill investment are themselves unequally distributed. The relative cost of acquiring a bridge language is partly inherited, and language behavior varies with these inherited linguistic conditions. Formal English-medium schooling is strongly organized by state boundaries, particularly in the public sector. Within the same district, social-group differences remain large in English-medium and private-school participation, with larger differences in urban schooling markets. EMI exposure is more strongly associated with later local consumption than enrollment alone, and this association is stronger in districts with higher initial human capital and urbanization. The inherited linguistic gradient is also associated with later changes in human capital, the skill composition of recent work migrants, financial access, and sectoral composition.

Taken together, the evidence limits the equalizing promise of a high-return educational opportunity. Households begin from unequal linguistic conditions and face unequal access to EMI; among districts, the later consumption association is stronger where baseline human capital and urbanization are higher. English skills can be valuable while access to English-medium schooling and the local conditions associated with its later economic returns remain unequal.

4 Access to English-Medium Instruction

4.1 Measuring Access to English-Medium Schooling

Inherited linguistic conditions can change the relative cost of acquiring English, but households can invest only through schooling options they can reach. I therefore use the measures defined in Section 2 to describe realized access to English-medium schooling. The NSS 64th Round records household choices in 2007–08: whether children ages 5–19 are enrolled, the reported medium of instruction among enrolled children, and school management among enrolled children with reported management (National Sample Survey Office 2011). These outcomes jointly reflect available schools, household preferences, and household resources.

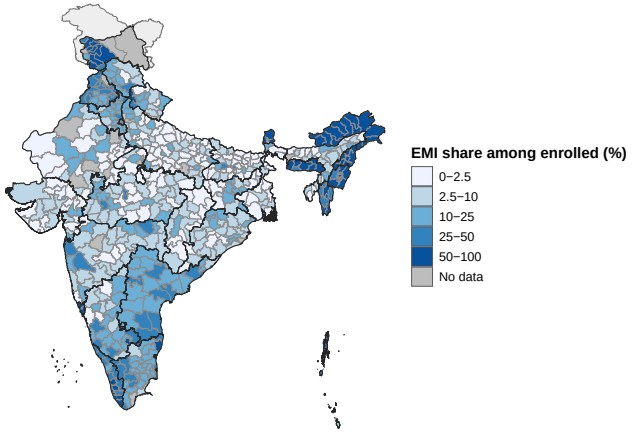
The schooling measures capture different components of realized access. Enrollment records whether an age-eligible child attends school. EMI among enrolled children records medium of instruction among children already

in school, and EMI exposure among all children incorporates both enrollment and reported medium. I also separate public- and private-school EMI exposure and private-school enrollment. School management and medium of instruction identify different schooling choices: households choosing private schools often cite perceived learning quality or school environment more often than English medium itself (Joshi and Kumar 2017). I therefore keep enrollment, EMI among enrolled children, EMI exposure among all children, public EMI, private EMI, and private enrollment separate.

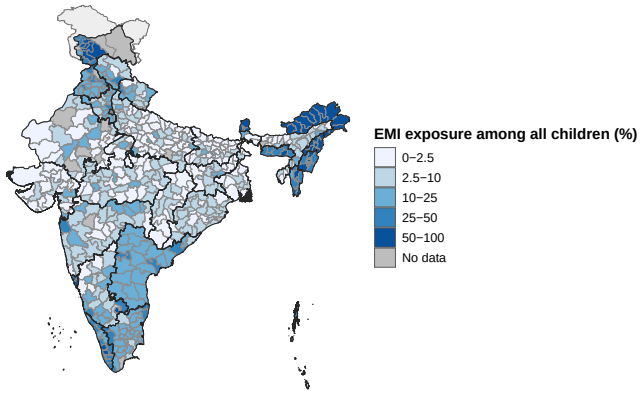
Reported medium of instruction is distinct from English proficiency and instructional quality. In the rural sample studied by Lahoti and Mukhopadhyay (2019), 57% of children described as attending English-medium schools were actually taught in the dominant regional language, while another 18% used English books with explanation in the regional language. The NSS and DISE measures therefore record reported school medium; they do not measure English proficiency or classroom language use.

DISE reports district enrollment by medium and management from schools. These enrollment shares reflect both available provision and family demand. I use DISE to corroborate the geographic pattern measured in the NSS because the two datasets record the same schooling margin from different respondents (Mehta 2022; District Information System for Education 2017).

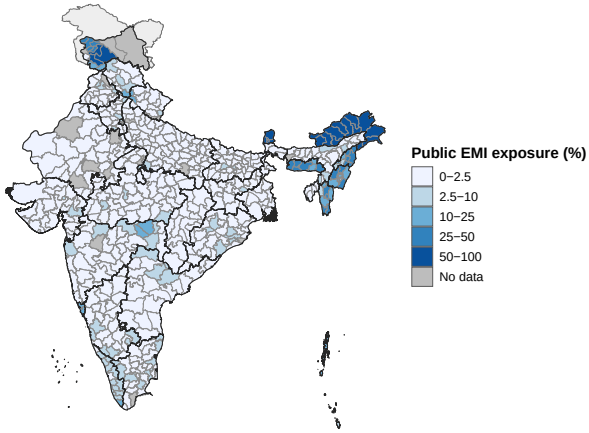
Figure 3 compares the four EMI measures directly. EMI among enrolled children isolates the medium-of-instruction margin among children already in school, while EMI exposure among all children also incorporates enrollment. Splitting exposure among all children by school management shows that public-school EMI is concentrated especially in northern and northeastern districts, whereas private-school EMI is more geographically dispersed. The four panels use the same percentage bins so that differences in both levels and geography can be compared directly.



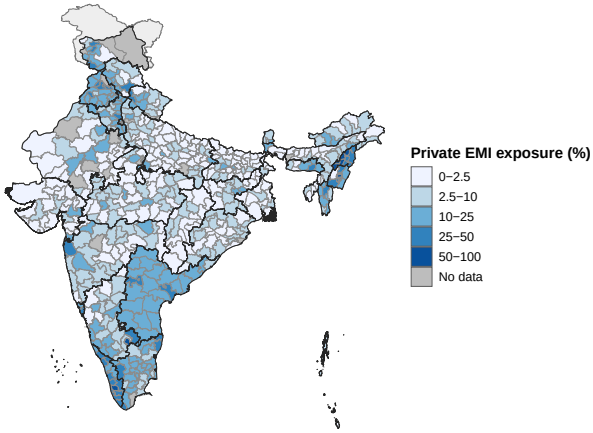
(a) EMI share among enrolled children.



(b) EMI exposure among all children.



(c) Public-school EMI exposure among all children.



(d) Private-school EMI exposure among all children.

Figure 3: District distribution of English-medium schooling measures on 2001 Census boundaries. All panels use a common percentage scale. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011). Tengshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.

4.2 EMI Access as a Function of Institutional Factors

The schooling measures are organized very differently across states. Table 3 presents the same three geographic specifications for each schooling outcome: an unadjusted association, a specification with region fixed effects and predetermined 2001 Census controls, and a specification with state fixed effects and the same controls. The table reports standardized coefficients so that attenuation can be compared across outcomes, with state-clustered standard errors beneath each estimate. Nationally, linguistic distance is positively associated with every English-

medium schooling margin in the table. The association attenuates sharply within states: the standardized coefficient with state fixed effects is about 0.12 for enrollment, 0.03 for EMI among enrolled children, 0.06 for EMI exposure among all children, 0.05 for public EMI, 0.09 for private EMI, and 0.05 for DISE English-medium enrollment.

The state membership R^2 rows report the share of district variation associated with state indicators. State indicators explain roughly 84% of linguistic-distance variation. Across the schooling measures, they explain about 83–84% of EMI exposure and EMI among enrolled children, about 83% of DISE English-medium enrollment, and nearly 90% of public EMI, compared with about 57% of private EMI and 36% of enrollment. The especially strong state concentration of public EMI accords with the historical organization of Indian state boundaries and schooling systems around language. Jain (2017) shows that aligning official instructional language with local language after linguistic state reorganization substantially improved educational attainment.

Table 3: Linguistic Distance and English-Medium Schooling across Geographic Specifications

	(1)	(2)	(3)
Enrollment			
Linguistic distance from Hindi	0.249***	0.177**	0.120
	(0.082)	(0.071)	(0.079)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.359	0.359	0.359
English-medium share among enrolled			
Linguistic distance from Hindi	0.364***	0.175	0.033
	(0.113)	(0.132)	(0.069)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.840	0.840	0.840
English-medium exposure among all children			
Linguistic distance from Hindi	0.376***	0.190	0.063
	(0.112)	(0.134)	(0.082)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.829	0.829	0.829
Public-school English-medium exposure			
Linguistic distance from Hindi	0.331**	0.176	0.055
	(0.130)	(0.126)	(0.084)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes

Table 3: Linguistic Distance and English-Medium Schooling across Geographic Specifications *(continued)*

	(1)	(2)	(3)
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.895	0.895	0.895
Private-school English-medium exposure			
Linguistic distance from Hindi	0.450***	0.190*	0.093
	(0.089)	(0.085)	(0.059)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	568	568	568
State membership R^2	0.570	0.570	0.570
DISE English-medium enrollment share			
Linguistic distance from Hindi	0.403***	0.142	0.045
	(0.112)	(0.109)	(0.057)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	520	520	520
State membership R^2	0.841	0.841	0.841

Each outcome block reports standardized coefficients from district regressions on speaker-weighted linguistic distance from Hindi. State-clustered standard errors are in parentheses. Columns (1)-(3) are unadjusted, region fixed effects plus baseline 2001 Census covariates, and state fixed effects plus the same covariates, respectively. Each outcome uses one fixed complete-case sample across its three specifications. State membership R^2 comes from a regression of each outcome on state indicators alone, using the districts available for that outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sources: NSS 64th Round, 2007–08; DISE District Report Cards, 2007–08; Census of India 2001.

DISE corroborates the NSS geographic pattern. DISE English-medium enrollment and NSS EMI among enrolled children have a raw district correlation of 0.896, while DISE and NSS EMI exposure among all children correlate at 0.882. After subtracting each state's mean from both measures, the correlations are 0.607 and 0.562. These within-state correlations remain substantial even after removing the broad state component that organizes both schooling measures. The corresponding district-level NSS–DISE comparison is shown in Figure 7.

Private schooling has a distinct household cost. Even schools described as low-fee can absorb a substantial share of poor households' income (Endow 2019), and private enrollment is strongly selected on household characteristics (Muralidharan and Kremer 2008; Raju 2023). Raw public-private achievement differences therefore do not identify differences in school effectiveness. In a randomized voucher experiment in Andhra Pradesh, Muralidharan and Sundararaman (2015) find no statistically significant difference between lottery winners and losers in Telugu, mathematics, English, or science and social studies after two and four years, despite large cross-sectional differences between public- and private-school students.

School management identifies whether enrollment is in the public or private sector. I analyze the DISE measures

of pupil-teacher ratios, single-teacher schools, sanitation, private-sector composition, and EMI separately.

4.3 EMI Access as a Function of Social Factors

Access differs sharply across social groups even within the same districts. Figure 4 compares Scheduled Tribe (ST), Scheduled Caste (SC), and Other Backward Class (OBC) children with children in the residual Other category within common districts. The social-group differences are present for enrollment but substantially larger for English-medium and private schooling. Relative to Other children, mean district enrollment is lower by about 13.6 percentage points for ST children, 9.8 points for SC children, and 6.4 points for OBC children. EMI among enrolled children is lower by about 13.3, 17.2, and 13.2 points respectively. Private enrollment is lower by about 18.1 points for ST children, 23.5 for SC children, and 14.5 for OBC children. Household resources and social institutions can shape these schooling choices (Endow 2019; Raju 2023). In Mumbai, Munshi and Rosenzweig (2006) show that caste networks continued to direct lower-caste boys toward local-language schools associated with traditional occupations as returns to nontraditional white-collar work rose, while lower-caste girls shifted more rapidly toward English schools.

The rural/urban comparison makes the social stratification more pronounced. For SC children, for example, the difference in EMI among enrolled children is about 8.4 percentage points in rural districts and 24.6 points in urban districts; the corresponding difference in private EMI exposure is about 7.5 and 21.4 points. OBC and ST children show the same qualitative pattern. Urban schooling markets offer a wider range of formal options while household resources and social position continue to shape which options are reachable. This pattern describes sorting across households within urban districts.

The corresponding decompositions for boys and girls are similar. Government programs operating before the NSS round explicitly targeted girls' education (Press Information Bureau 2008), so I also examine the sex split directly. In this national cross-section, social group and rural/urban setting distinguish access more strongly than sex.

English-medium and private schools can be present in a district while remaining unequally accessible across households. The next section asks whether these observed EMI choices are associated with later economic outcomes that might help explain household demand for them.

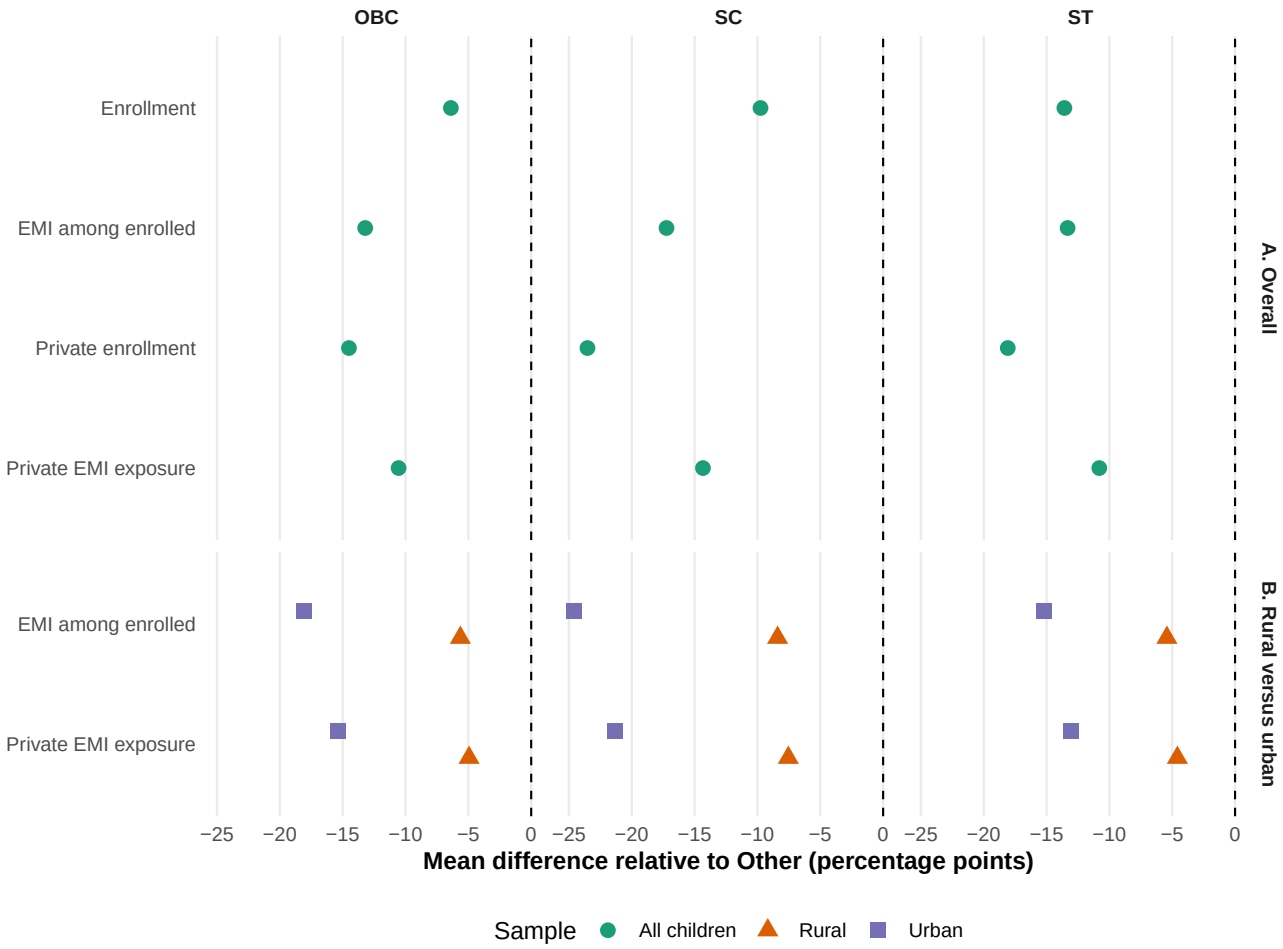


Figure 4: Social-group differences in access to English-medium and private schooling. Panel A reports mean within-district percentage-point differences for OBC, SC, and ST children relative to the Other category in enrollment, EMI among enrolled children, private enrollment, and private EMI exposure among all children. Panel B separates rural and urban differences for EMI among enrolled children and private EMI exposure. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011).